DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

B.Tech (AI & ML) Academic Assignment

WEB DEVELOPMENT PROJECT REPORT

A Comprehensive Study on Front-End Web Design using HTML5 and CSS3

STUDENT SUBMISSION DETAILS

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Section:	CSM3
Subject:	Web Development
Technologies:	HTML5 & CSS3
Academic Year:	First Year B.Tech (2026)

Submitted for Academic Assessment in Web Development Course

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1. Introduction to Web Development

Web development is the process of creating and maintaining websites and web applications that run online on web browsers. It involves front-end design, back-end server management, and database configuration. In modern computer science education, understanding front-end web design is fundamental.

This report details three practical web development assignments completed using core client-side technologies. Front-end design focuses directly on user experience, visual layouts, colors, responsive structure, and interactive components.

2. Technologies Used

Technology	Role in Web Development	Key Features Applied
HTML5 (HyperText Markup Language)	Provides the structural skeleton and layout of the webpage content.	Semantic tags (`<header>`, `<section>`, `<footer>`), input forms, image containers, headings, and lists.
CSS3 (Cascading Style Sheets)	Handles visual presentation, color styling, layout alignments, and animations.	Flexbox, CSS Grid, custom card designs, hover effects, background gradients, and typography control.

Project 1: Personal Portfolio Website

1.1 Introduction

A personal portfolio website is an essential tool for engineering students to showcase their academic profiles, educational background, technical skills, personal hobbies, completed projects, achievements, and contact details in a single online hub.

1.2 Aim

To design and construct a responsive Personal Portfolio Website using HTML5 for structure and CSS3 for visual styling.

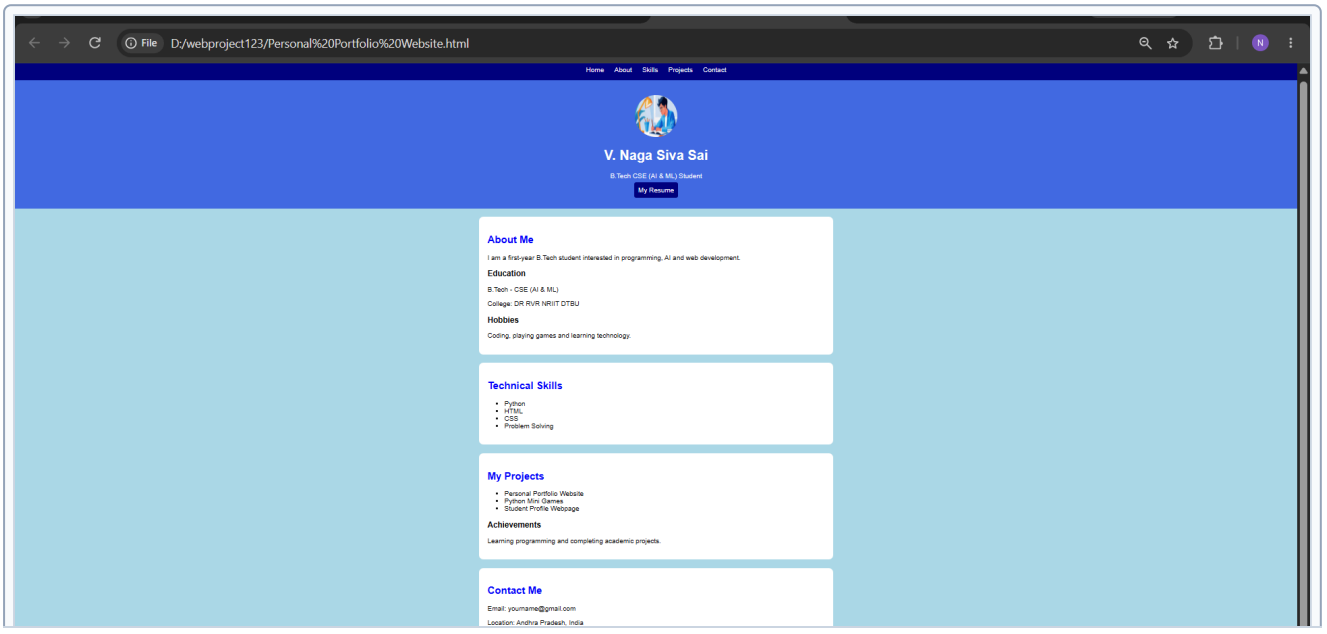
1.3 Key Features

- **Navigation Menu:** Smooth navigation links (`Home`, `About`, `Skills`, `Projects`, `Contact`).
- **Profile Banner:** Displays student avatar, name, current program, and a quick link button to the resume.
- **Educational Qualifications:** Structured sections detailing degree, branch, and college name.
- **Skills & Hobbies:** Categorized bullet points for programming skills (Python, HTML, CSS) and hobbies.
- **Projects & Achievements:** Lists completed mini-projects and academic accomplishments.
- **Contact Section:** Direct contact details with location and email links styled using CSS buttons.

1.4 Usage Instructions

To view this webpage, open the file `Personal Portfolio Website.html` directly in any standard browser (such as Google Chrome, Microsoft Edge, or Firefox). Alternatively, right-click the file inside VS Code and choose **"Open with Live Server"** for live browser rendering.

1.5 Project Output



Project 2: User Registration and Login Interface

2.1 Introduction

The User Registration and Login Interface provides a front-end setup for web applications that require user authentication, user account setup, and password recovery workflows.

2.2 Aim

To design a clean, responsive set of user account management forms using HTML5 form controls and CSS3 card styling.

2.3 Key Features

- **Registration Form:** Form fields for Full Name, Email, Password, and Confirm Password with submit button.
- **Login Form:** Direct authentication input fields with a "Forgot Password?" recovery link.
- **Password Recovery Form:** Clean input card allowing users to enter their registered email address.
- **Profile Setup Form:** Secondary form with input controls for Display Name, Phone Number, and About Me text area.
- **Form Validations & UX:** Built using HTML validation attributes (`required`, input types) and custom CSS hover and focus highlights.

2.4 Usage Instructions & Validation

Users can interact with the form fields directly. Browser-level validation enforces correct email formatting and required field population prior to submission. *Note: Actual user authentication and database storage require a backend server engine (e.g., Node.js or Python) or JavaScript integrations.*

2.5 Project Output



Output 2: User Registration, Login, Password Recovery, and Profile Setup Forms

Project 3: Restaurant and Cafe Website

3.1 Introduction

The Restaurant and Cafe Website presents an online business presence for "RK Restaurant & Cafe", highlighting menus, chef credentials, testimonials, hours of operation, and table bookings.

3.2 Aim

To build an interactive, thematic, multi-section restaurant website utilizing modern CSS layouts and responsive cards.

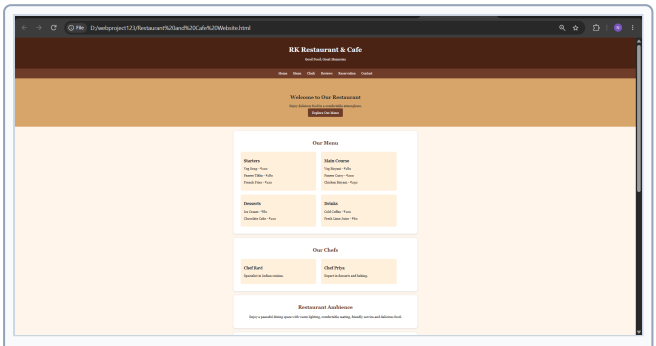
3.3 Key Features

- **Brand Header & Welcome Section:** Warm thematic banner featuring business branding and quick navigation links.
- **Categorized Food Menu:** Organized grid cards for Starters, Main Course, Desserts, and Drinks with pricing in INR (₹).
- **Chef Profiles & Testimonials:** Highlighted cards for head chefs and real customer reviews.
- **Table Reservation Form:** Date, time, guest count, and contact input controls for online booking requests.
- **Location & Hours:** Clear display of weekly opening hours and restaurant contact details.

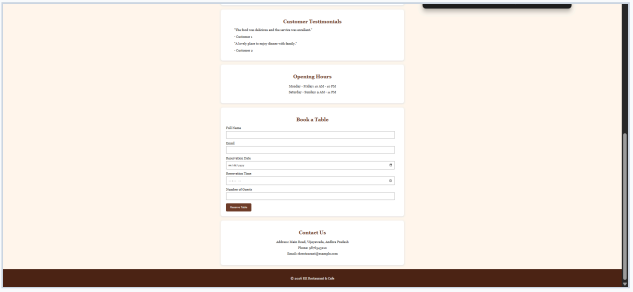
3.4 Usage Instructions

Visitors can scroll smoothly through the single-page web app to review dishes, check operational hours, and submit reservation requests through the sample form.

3.5 Project Outputs



Output 3.1: Menu & Chef Profiles



Output 3.2: Reservation & Reviews

6. Conclusion & Key Learning Outcomes

Through the completion of these three front-end web development projects, I gained foundational experience in constructing web applications using HTML5 and CSS3. Building diverse project types—from personal portfolios to interactive business websites—helped strengthen my understanding of core web design principles.

Summary of Skills Acquired

- **Semantic HTML5 Structure:** Mastered the use of content-specific markup tags (`<header>`, `<section>`, `<article>`, `<footer>`) to build structured pages.
- **CSS Layout Techniques:** Applied CSS Flexbox and Grid containers to align content panels, navigation bars, and form elements.
- **Form Design & Front-End Validation:** Created complex form layouts utilizing standard input attributes (`required`, `type="email"`, `type="date"`) and accessible labels.
- **UI/UX Enhancement:** Implemented visual hierarchy, hover transitions, custom button styles, and color schemes suited to specific domains.
- **Project Workflow:** Practiced managing local source code files and rendering webpages locally using browser developer tools and VS Code Live Server.

Student Declaration

I hereby declare that the web development projects presented in this report were developed as part of my first-year B.Tech curriculum coursework. All outputs and screenshots included represent genuine work completed using HTML5 and CSS3.

Date: October 24, 2026
Location: Andhra Pradesh, India

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